

9. Stationary Points and Extrema

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Stationary Points

Definition

Let $\Omega \subseteq \mathbb{R}^n$, $f : \Omega \rightarrow \mathbb{R}$ and $\mathbf{a} \in \Omega$. Then

- $\mathbf{a}$ is an **absolute** or **global maximum** of f if $f(\mathbf{a}) \geq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$;
- $\mathbf{a}$ is an **absolute** or **global minimum** of f if $f(\mathbf{a}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$;
- $\mathbf{a}$ is a **local maximum** of f if there is an open set $U_{\mathbf{a}} \ni \mathbf{a}$ such that $f(\mathbf{a}) \geq f(\mathbf{x})$ for all $\mathbf{x} \in U_{\mathbf{a}}$;
- $\mathbf{a}$ is a **local minimum** of f if there is an open set $U_{\mathbf{a}} \ni \mathbf{a}$ such that $f(\mathbf{a}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in U_{\mathbf{a}}$;
- $\mathbf{a}$ is a **stationary point** of f if f is differentiable at $\mathbf{a}$ and $\nabla f(\mathbf{a}) = \underline{0}$;
- $\mathbf{a}$ is a **saddle point** of f if it is a stationary point of f but is neither a local maximum nor a local minimum.

Stationary Points

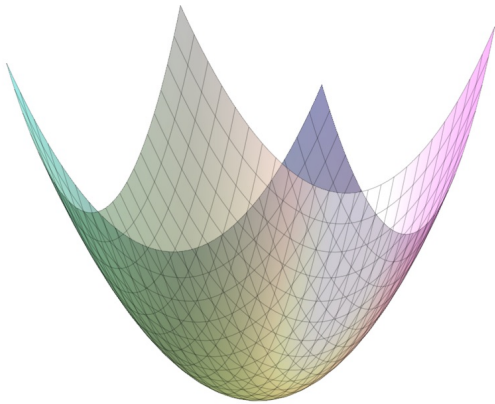


Figure: A local minimum

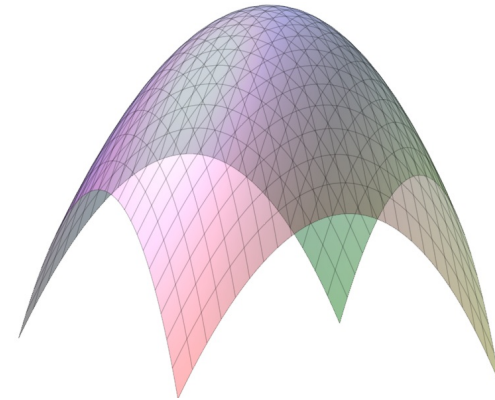


Figure: A local maximum

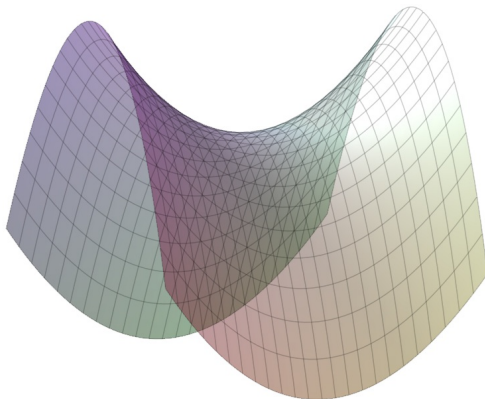


Figure: A saddle point

$$x^2 + y^2 \sin\left(\frac{1}{x^2 + y^2}\right)$$

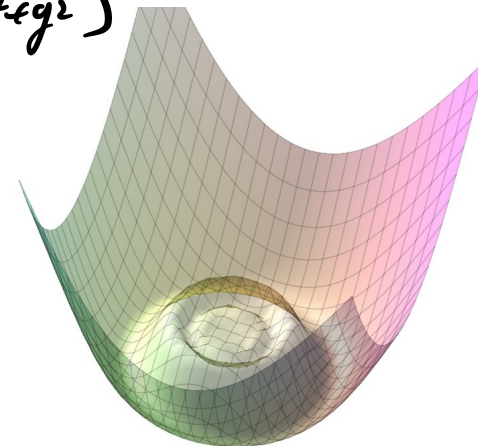


Figure: A 'saddle' point

Critical Points

The following theorem is a generalisation of the MATH1131 theorem regarding 'critical points', and its proof is similar to the proof given in MATH1131.

Theorem

Suppose that $\mathbf{a} \in \mathbb{R}^n$ is a local maximum or minimum of a function $f : \Omega \rightarrow \mathbb{R}$. Then either

- (1) $\mathbf{a}$ is a stationary point; or
- (2) $\mathbf{a} \in \partial\Omega$; or
- (3) f is not differentiable at $\mathbf{a}$. $(\mathbf{a} \in \text{Int}(\Omega))$

Points satisfying (1), (2) or (3) are called **critical points of f** .

Critical Points

Proof sketch.

WLOG, suppose $\underline{a}$ is a local max of f .

Suppose (2) and (3) are not true, then $\underline{a}$ is an interior point of Ω , so there is $\varepsilon > 0$ st.

$B(\underline{a}, \varepsilon) \subseteq \Omega$, with the property that $f(\underline{x}) \leq f(\underline{a})$ for all $\underline{x} \in B(\underline{a}, \varepsilon)$. For each $1 \leq j \leq n$, define

$$g_j : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R} \text{ by} \\ g_j(t) = f(\underline{a} + t\underline{e}_j).$$

Then g_j has a local max at $t=0$, and so

$$0 = g_j'(0) = \lim_{h \rightarrow 0} \frac{g_j(h) - g_j(0)}{h} = \lim_{h \rightarrow 0} \frac{f(\underline{a} + h\underline{e}_j) - f(\underline{a})}{h} = \frac{\partial f}{\partial x_j}(\underline{a})$$

This holds for all j , so $\nabla f(\underline{a}) = \underline{0}$, and so $\underline{a}$ is a stationary point of f .

Critical Points

Exercise. Find all critical points of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = xy(x + y)$.

First note that f is differentiable everywhere
and $\partial(\mathbb{R}^2) = \emptyset$, so we only look for stationary pts.

$$f(x, y) = x^2y + xy^2$$

$$\underline{0} = \nabla f = \begin{pmatrix} 2xy + y^2 \\ 2xy + x^2 \end{pmatrix} \Rightarrow \begin{cases} 2xy + y^2 = 0 & (1) \\ 2xy + x^2 = 0 & (2) \end{cases}$$

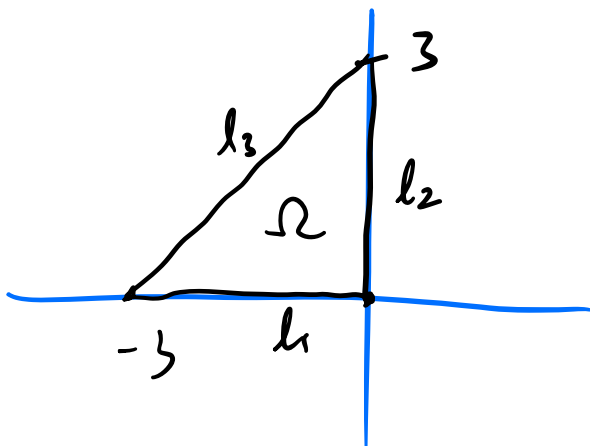
Can see that $x=y=0$ is a solution, but are there more?

Then $(1) - (2)$ gives $x^2 = y^2$, so $x = \pm y$.

In both cases, substituting into (1) gives $y^2 = 0$
and so $(0, 0)$ is the only critical point of f .

Critical Points

Exercise. Find the maximum and minimum values of $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = x^3 - y^3 - 3xy$ on the region Ω bounded by the lines $x = 0$, $y = 0$ and $y = x + 3$.



$$\begin{aligned} l_1 : \underline{c}_1 : [0, 3] &\rightarrow \mathbb{R}^2, \quad \underline{c}_1(t) = (-t, 0) \\ l_2 : \underline{c}_2 : [0, 3] &\rightarrow \mathbb{R}^2, \quad \underline{c}_2(t) = (0, t) \\ l_3 : \underline{c}_3 : [0, 3] &\rightarrow \mathbb{R}^2, \quad \underline{c}_3(t) = (-t, 3-t) \end{aligned}$$

For l_1 : $f(\underline{c}_1(t)) = -t^3$, which has max of 0 and min of -27

For l_2 : $f(\underline{c}_2(t)) = -t^3$, same max and min

$$\begin{aligned} \text{For } l_3 : f(\underline{c}_3(t)) &= -t^3 - (3-t)^3 + 3(t)(3-t) \\ &= -12t^2 + 36t - 27 \end{aligned}$$

Critical Points

Exercise cont. Find the maximum and minimum values of $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = x^3 - y^3 - 3xy$ on the region Ω bounded by the lines $x = 0$, $y = 0$ and $y = x + 3$. *closed region*

$$f\left(\frac{1}{2}, \frac{3}{2}\right) = -3(4t^2 - 12t + 9) = -3(2t - 3)^2$$

$$\text{which has max } f\left(\frac{1}{2}\right) = 0 \text{ and min } f(0) = -27$$

Check stationary points:

$$\nabla f = \begin{pmatrix} 3x^2 - 3y \\ -3y^2 - 3x \end{pmatrix} = \begin{cases} y = x^2 \\ -x = y^2 \end{cases}$$

Sub 2nd equation into 1st equation: $y = (-y^2)^2 = y^4$

$$\Rightarrow 0 = y^4 - y = y(y^3 - 1) = y(y - 1)(y^2 + y + 1)$$

$$\Rightarrow y = 0, 1 \Rightarrow x = 0, -1$$

$$f(-1, 1) = (-1)^3 - 1^3 - 3(-1)(1) = 1$$

So max of f is 1 at $(-1, 1)$ and the min of f is -27 at $(0, 3)$ and $(-3, 0)$

Classifying Stationary Points

H is symmetric

a is interior

Suppose that $\mathbf{a}$ is a stationary point of a C^2 function $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. By Taylor's Theorem (slightly more general than shown in the previous chapter), there is a function $R_{2,\mathbf{a}}$ such that $\frac{R_{2,\mathbf{a}}(\mathbf{x})}{|\mathbf{x} - \mathbf{a}|^2} \rightarrow 0$ as $\mathbf{x} \rightarrow \mathbf{a}$ and

$$f(\mathbf{x}) = f(\mathbf{a}) + \underbrace{\nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a})}_{=0} + \frac{1}{2} \left((\mathbf{x} - \mathbf{a}) H_f(\mathbf{a}) (\mathbf{x} - \mathbf{a})^T \right) + R_{2,\mathbf{a}}(\mathbf{x}).$$

Since $H_f(\mathbf{a})$ is real and symmetric, its eigenvalues are real and $\mathbb{R}^n$ has an orthonormal basis of eigenvectors of $H_f(\mathbf{a})$ (MATH2501/2601). In the case of $\mathbf{x} - \mathbf{a}$ being an eigenvector of $H_f(\mathbf{a})$ with eigenvalue λ and $|\mathbf{x} - \mathbf{a}|$ small, we get that

$$f(\mathbf{x}) \sim f(\mathbf{a}) + \frac{1}{2} \lambda |\mathbf{x} - \mathbf{a}|^2.$$

full rank if $\lambda \neq 0$

First suppose none of the eigenvalues are 0. If all the eigenvalues are positive, then $\mathbf{a}$ is a local minimum. If all the eigenvalues are negative, then $\mathbf{a}$ is a local max. Otherwise $\mathbf{a}$ is a saddle point.

If 0 is an eigenvalue, we need more information.

Classifying Stationary Points

Definition

A symmetric matrix is said to be

- **positive definite** if all of its eigenvalues are positive;
- **positive semi-definite** if all of its eigenvalues are non-negative;
- **negative definite** if all of its eigenvalues are negative;
- **negative semi-definite** if all of its eigenvalues are non-positive.

Theorem (Second Derivative Test)

Let $\Omega \subseteq \mathbb{R}^n$ be open and $f \in C^2(\Omega)$, and suppose that $\mathbf{a} \in \Omega$ is a stationary point of f .

- If $H_f(\mathbf{a})$ is positive definite, then $\mathbf{a}$ is a local minimum of f ;
- If $H_f(\mathbf{a})$ is negative definite, then $\mathbf{a}$ is a local maximum of f ;
- If $\mathbf{a}$ is a local minimum of f , then $H_f(\mathbf{a})$ is positive semi-definite;
- If $\mathbf{a}$ is a local maximum of f , then $H_f(\mathbf{a})$ is negative semi-definite.

Classifying Stationary Points

Exercise. Find and classify all stationary points of $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x^3 + 6x^2 + 3y^2 - 12xy + 9x.$$

$$\underline{0} = \nabla f = \begin{pmatrix} 3x^2 + 12x - 12y + 9 \\ 6y - 12x \end{pmatrix}$$

$$\begin{aligned} \text{So } y = 2x, \text{ and } 0 &= 3x^2 + 12x - 12(2x) + 9 \\ &= 3x^2 - 12x + 9 \\ &= 3(x^2 - 4x + 3) = 3(x-1)(x-3) \end{aligned}$$

So $x=1, 3$ and $y=2, 6$ respectively

$$f_{xx} = 6x + 12$$

$$f_{yy} = 6$$

$$f_{xy} = -12$$

Classifying Stationary Points

Exercise cont. Find and classify all stationary points of $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x^3 + 6x^2 + 3y^2 - 12xy + 9x.$$

$$\mu_1 = H_f(1, 2) = \begin{pmatrix} 18 & -12 \\ -12 & 6 \end{pmatrix}$$

$$\det(\mu_1 - \lambda I) = 0$$
$$= \det \begin{pmatrix} 18-\lambda & -12 \\ -12 & 6-\lambda \end{pmatrix}$$

$$= (18-\lambda)(6-\lambda) - 144$$

$$= \lambda^2 - 24\lambda - 36$$

$$\lambda = \frac{24 \pm \sqrt{24^2 - 4(-36)}}{2}$$

$$= \frac{24 \pm \sqrt{720}}{2} = 12 \pm \sqrt{180}$$

$(1, 2)$ is a Saddle point

$$\mu_2 = H_f(3, 6) = \begin{pmatrix} 30 & -12 \\ -12 & 6 \end{pmatrix}$$

$$\det(\mu_2 - \lambda I) = \det \begin{pmatrix} 30-\lambda & -12 \\ -12 & 6-\lambda \end{pmatrix}$$
$$= (30-\lambda)(6-\lambda) - 144$$

$$= \lambda^2 - 36\lambda + 36$$

$$\lambda = \frac{36 \pm \sqrt{36^2 - 4(36)}}{2}$$

$$= \frac{36 \pm \sqrt{1152}}{2}$$

$$= 18 \pm \sqrt{288}$$

$(3, 6)$ is a local min

can check
sum of
roots are
positive so
does product.



Sylvester's Criterion

$$\begin{pmatrix} H_{11} & \dots & H_{1n} \\ \vdots & & \vdots \\ H_{n1} & \dots & H_{nn} \end{pmatrix}$$

Theorem (Sylvester's Criterion)

Let $H = (H_{ij})_{i,j=1}^n$ be a real symmetric matrix, H_k the upper $k \times k$ submatrix of H defined by $(H_k)_{ij} = H_{ij}$ for all $1 \leq i, j \leq k$, and $\Delta_k = \det(H_k)$. Then

- H is positive definite if and only if $\Delta_k > 0$ for all k ;
- H is positive semi-definite implies $\Delta_k \geq 0$ for all k ;
- H is negative definite if and only if $(-1)^k \Delta_k > 0$ for all k ;
- H is negative semi-definite implies $(-1)^k \Delta_k \geq 0$ for all k .

Corollary (Alternative Second Derivative Test)

Suppose that $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a C^2 function and $\nabla f(\mathbf{a}) = 0$. If $\det(H_f(\mathbf{a})) < 0$, then $\mathbf{a}$ is a saddle point. Otherwise, if $\det(H_f(\mathbf{a})) > 0$, then $\mathbf{a}$ is

- a local min if $f_{xx}(\mathbf{a}) > 0$;
- a local max if $f_{xx}(\mathbf{a}) < 0$.

The test is inconclusive when $\det(H_f(\mathbf{a})) = 0$.

Sylvester's Criterion

Example. Recall the function from earlier defined by

$$f(x, y) = x^3 + 6x^2 + 3y^2 - 12xy + 9x.$$

$$H_f(1, 2) = \begin{pmatrix} 18 & -12 \\ -12 & 6 \end{pmatrix}$$

$$\det(H_f(1, 2))$$

$$= 18 \times 6 - 144$$

$$= 108 - 144 = -36 < 0$$

so $(1, 2)$ is a
saddle point

$$H_f(3, 6) = \begin{pmatrix} 30 & -12 \\ -12 & 6 \end{pmatrix}$$

$$\det(H_f(3, 6)) = 30 \times 6 - 144 \\ = 36$$

since $f_{xx}(3, 6) = 30 > 0$,
so $(3, 6)$ is a local
min.

Extrema Subject to Constraint(s)

What if we want to find the local minima or maxima of a function but over a subset of the function's domain?

For example, the yield of a gas-powered machine $Y(P, T, V)$ is dependent on the pressure P , temperature T and volume V of the gas, but the gas must always satisfy the equation $PV = nRT$ for some constants n and R .

Another example could be to find the point on the curve $x^2 + xy - y^2 = 2$ that is closest to the origin, where we minimise the function $(x, y) \mapsto x^2 + y^2$ with the constraint equation being the equation of the curve.

More generally, we could want to find the extrema of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ subject to the constraint $g(\mathbf{x}) = C$, where C is a constant and $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable.

f, g

Extrema Subject to Constraint(s)

Let $S = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = C\}$ and suppose that $\mathbf{a}$ is a local extremum of $f|_S$ (the restriction of f to S).

consider a differentiable curve $\underline{c} : I \rightarrow S$
with $0 \in \text{Int}(I)$ and $\underline{c}(0) = \underline{a}$. Then

$f \circ \underline{c} : I \rightarrow \mathbb{R}$ has a local extremum at 0.

As $f \circ \underline{c}$ is differentiable, 0 is a stationary point of $f \circ \underline{c}$:

$$\begin{aligned} 0 &= (f \circ \underline{c})'(0) = Df(\underline{c}(0)) \underline{c}'(0) && \text{(Chain Rule)} \\ &= \nabla f(\underline{c}(0)) \cdot \underline{c}'(0) \\ &= \nabla f(\underline{a}) \cdot \underline{c}'(0) \end{aligned}$$

As $\nabla f(\underline{a})$ is $\perp$ to all tangent vectors to S at $\underline{a}$,
it is normal to S at $\underline{a}$. As $\nabla g(\underline{a})$ is also
normal to S at $\underline{a}$, Then $\nabla f(\underline{a}) \parallel \nabla g(\underline{a})$.

Method of Lagrange Multipliers

Theorem

Let $f, g \in C^1(\mathbb{R}^n)$ for some $n \geq 2$ and $\mathcal{S} = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = C\}$ for some $C \in \mathbb{R}$ such that $\mathcal{S}$ defines a surface. If $\mathbf{a} \in \mathcal{S}$ is a local minimum or maximum of f on $\mathcal{S}$ and $\nabla g(\mathbf{a}) \neq \mathbf{0}$, then there exists $\lambda \in \mathbb{R}$, the **Lagrange Multiplier**, such that

$$\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a}).$$

Remarks.

- Recall that $\nabla g(\mathbf{a}) \neq \mathbf{0} \iff \frac{\partial g}{\partial x_i}(\mathbf{a}) \neq 0$ **for some** $i \in \{1, \dots, n\}$; and
- $\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a}) \iff \frac{\partial f}{\partial x_i}(\mathbf{a}) = \lambda \frac{\partial g}{\partial x_i}(\mathbf{a})$ **for all** $i \in \{1, \dots, n\}$.

Method of Lagrange Multipliers

Exercise. Find the minimum and maximum values of $f(x, y) = 2x + 3y$ subject to $g(x, y) = x^2 + y^2 = 4$ and the points at which they occur.

We can observe that the circle of radius 2 is compact, and so a global max and min for the restriction of f are guaranteed.

$$\nabla f = \lambda \nabla g \Leftrightarrow \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \lambda \begin{pmatrix} 2x \\ 2y \end{pmatrix} \quad \text{Note here that } \lambda \neq 0 \text{ and } \underline{x} \neq \underline{0}$$

$$1 = \lambda x \rightarrow x = \frac{1}{\lambda}$$

$$3 = 2\lambda y \rightarrow y = \frac{3}{2\lambda}$$

$$x^2 + y^2 = 4 \rightarrow \left(\frac{1}{\lambda}\right)^2 + \left(\frac{3}{2\lambda}\right)^2 = 4 \Rightarrow \frac{1}{\lambda^2} + \frac{9}{4\lambda^2} = 4$$
$$\Rightarrow 4 + 9 = 16\lambda^2$$
$$\Rightarrow \lambda = \pm \frac{\sqrt{13}}{4}$$

Often, you will want to argue that $\lambda \neq 0$ or $x \neq 0$.

Method of Lagrange Multipliers

$$\lambda = \pm \frac{\sqrt{13}}{4} \Rightarrow x = \pm \frac{4}{\sqrt{13}}$$

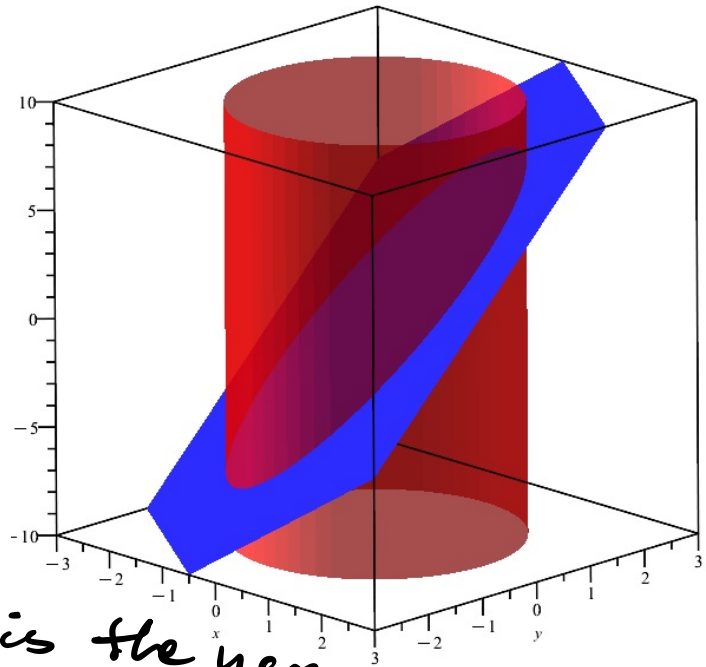
$$\Rightarrow y = \pm \frac{3}{2 \frac{\sqrt{13}}{4}} = \pm \frac{6}{\sqrt{13}}$$

So the extrema occur at

$$\pm \left(\frac{4}{\sqrt{13}}, \frac{6}{\sqrt{13}} \right).$$

$$f\left(\frac{4}{\sqrt{13}}, \frac{6}{\sqrt{13}}\right) = \frac{8}{\sqrt{13}} + \frac{18}{\sqrt{13}} = 2\sqrt{13}$$

$$f\left(-\frac{4}{\sqrt{13}}, -\frac{6}{\sqrt{13}}\right) = -2\sqrt{13} \quad \text{is the min}$$



Method of Lagrange Multipliers

$$d(\underline{x}, 0) = \sqrt{x^2 + y^2}$$

Exercise. Find the shortest distance from the curve defined by $x^2 + xy - y^2 = 2$ to the origin.

Want to extremise $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ $f(x, y) = x^2 + y^2$
subject to the $g(x, y) = x^2 + xy - y^2 = 2$.

$$\nabla f = \lambda \nabla g \Leftrightarrow \begin{pmatrix} 2x \\ 2y \end{pmatrix} = \lambda \begin{pmatrix} y + 2x \\ x - 2y \end{pmatrix}$$

$$2x = \lambda(y + 2x) \quad (1)$$

$$2y = \lambda(x - 2y) \quad (2)$$

$$x^2 + xy - y^2 = 2 \quad (3)$$

Note if $\lambda = 0 \Rightarrow \underline{x} = \underline{0}$, but $g(0, 0) \neq 2$.

Multiply (1) by y and (2) by x :

$$2xy = \lambda y(y + 2x) \quad (4)$$

$$2xy = \lambda x(x - 2y) \quad (5)$$

Method of Lagrange Multipliers

Ex. cont. Find the shortest distance from the curve defined by $x^2 + xy - y^2 = 2$ to the origin.

Equating (4) and (5) and dividing by λ gives

$$y(y+2x) = x(x-2y) \quad (6).$$

Expanding gives $y^2 + 2xy = x^2 - 2xy$ so

$$x^2 - 4xy - y^2 = 0 \quad \longrightarrow \quad (x-2y)^2 = 5y^2$$

Using (3), $x^2 - y^2 = 2 - xy$, and

$$2 - 5xy = 0, \quad \text{or} \quad \boxed{xy = \frac{2}{5}}$$

$y \neq 0,$

$$\text{so } \lambda = \pm \frac{2}{\sqrt{5}} \longleftarrow$$

$$\begin{aligned} &\downarrow \\ &\left(\frac{2y}{\lambda}\right)^2 = 5y^2 \\ &\frac{4y^2}{\lambda^2} = 5y^2 \end{aligned}$$

Method of Lagrange Multipliers

Ex. cont. Find the shortest distance from the curve defined by $x^2 + xy - y^2 = 2$ to the origin.

Multiplying (1) by x and (2) by y to get

$$2x^2 = \lambda (xy + 2x^2) \quad (6)$$

$$2y^2 = \lambda (xy - 2y^2) \quad (7)$$

Add (6) and (7), then cancel factor of 2:

$$f(x, y) = x^2 + y^2 = \lambda (x^2 + xy - y^2) = 2\lambda \text{ by (3)}$$

Since f restricted to the constraint is unbounded,
 $f(x, y) = 2\lambda = \frac{4}{\sqrt{5}}$ is a global minimum.

So the shortest distance from the curve to the origin is $\frac{2}{\sqrt{5}}$

Note: we did not need to find x or y